D A T A A N A L Y S T C A S E S T U D Y

BlinkIT

Grocery Sales & Outlet Performance Analytics

Company	BlinkIT	**Industry**	Quick-Commerce — Grocery Retail
Problem	Leadership can't tell which items, categories & outlet formats actually drive sales — assortment and expansion decisions run on guesswork.		
Tools	Excel | Power BI	**Difficulty**	Intermediate
Est. Time	8–10 hours		

1. Business Background

BlinkIT is a quick-commerce grocery platform operating 10 outlets across Tier 1, Tier 2, and Tier 3 cities, in formats ranging from small Grocery Stores to full Supermarket Type1/2/3 layouts. Outlets were established between 2011 and 2022, and the catalog spans 1,559 unique items across 16 categories such as Fruits and Vegetables, Snack Foods, Household, and Frozen Foods.

Category Management currently reviews sales by gut feel and outlet-manager anecdotes. There is no consolidated view of which items and outlet formats actually convert shelf space into sales, or whether newer outlets are catching up to established ones.

2. The Business Problem

The VP of Merchandising needs answers on: (1) which outlet formats and locations drive the most sales, (2) which item categories are over- or under-performing relative to shelf visibility, (3) whether item ratings actually predict sales, and (4) how outlet age affects performance. Without this, assortment planning and outlet-expansion budgets are being set without evidence.

Stakeholders: VP of Merchandising (assortment & category strategy) · Outlet Operations Lead (format & location performance) · Category Manager (item-level decisions) · You, Data & Analytics.

3. Your Role

You are the Data Analyst, reporting to the Head of Merchandising Analytics. You own the pipeline end-to-end: clean the raw export, resolve the data-quality issues below, answer stakeholder questions in SQL, and deliver a reusable Power BI dashboard. You're accountable for:

- Documented cleaning decisions
- Analytical accuracy (every dashboard number traceable to a query)
- Business framing (write for a VP, not a data scientist)
- A working — not static — dashboard

4. The Dataset

Single table **BlinkIT-Grocery-Data.csv** — one row per item-outlet combination, **8,523 rows × 12 columns**, pulled from the retail sales system.

Column	Type	Meaning
Item Identifier / Item Type	Text	Unique item code and its category (16 categories, e.g., Snack Foods, Dairy)
Item Fat Content	Text	Low Fat vs. Regular — inconsistently labeled (data defect, see below)
Item Weight / Item Visibility	Float	Product weight (kg); the shelf-space share the item occupies at that outlet (0–1 scale)
Outlet Identifier / Establishment Year	Text / Int	Which of the 10 outlets; year the outlet opened (2011–2022)
Outlet Location Type / Size / Type	Text	Tier 1–3 city; Small / Medium / High; Grocery Store or Supermarket Type1/2/3
Sales / Rating	Float	Sales value for that item at that outlet (USD); item rating on a 1–5 scale

Data Quality Issues

Data Quality Issue	Action Required
Item Fat Content has 5 raw labels for 2 categories: 'Low Fat' / 'LF' / 'low fat' and 'Regular' / 'reg'	Standardize casing and abbreviations into exactly two categories before any grouping — as-is it will fragment every category-level chart
Item Weight missing on 1,463 rows (17%)	Backfill from other rows of the same Item Identifier first, then fall back to a category median; document which rows remain unresolved
Item Visibility = 0 on 526 rows	A stocked product can't physically have zero shelf visibility — treat as a data artifact / missing value, not a literal 'invisible item'
Outlet Establishment Year ranges 2011–2022, including outlets opened as recently as 2022	State a reference year before computing Outlet Age; very new outlets will show near-zero history and can distort an 'age vs. sales' trend if not called out separately
Item Identifier repeats ~6,964 times across the 10 outlets (1,559 unique items)	This is the expected item-outlet grain, not a duplicate bug — don't accidentally deduplicate on Item Identifier alone and collapse the outlet dimension
Rating is recorded per item (1.0–5.0 in 0.1 steps)	Confirm what this rating measures (customer satisfaction vs. internal quality score) before building any 'rating drives sales' narrative

5. Task Breakdown — 3 Phases

Phase 1 — Excel (Cleaning & EDA)

Load & inspect nulls / dtypes → standardize Item Fat Content into two clean categories → impute Item Weight (same-item backfill, then category median) → treat zero Item Visibility as missing rather than literal → compute Outlet_Age from a stated reference year → produce 4+ labeled charts (sales by outlet type · top 10 item types by sales · visibility vs. sales scatter · sales by outlet size) → write 3+ business insights.

Phase 2 — Power BI (Dashboard)

Visuals: total sales KPI card · sales by outlet type bar · sales by location tier map/bar · item-type performance matrix · visibility-vs-sales scatter · outlet age vs. sales trend · rating distribution.

The dashboard must let a VP see, unaided: which outlet formats and locations perform best, where visibility is costing sales, and whether newer outlets are catching up.

6. Business Questions to Answer

The VP of Merchandising needs answers on:

(1) which outlet formats and locations drive the most sales,

(2) which item categories are over- or under-performing relative to shelf visibility,

(3) whether item ratings actually predict sales, and

(4) how outlet age affects performance. Without this, assortment planning and outlet-expansion budgets are being set without evidence.

7. Deliverables & Submission Format

- .xlsx — Excel workbook (Phase 1: cleaning + EDA + charts + written insights)
- .pbix — Power BI file: star-schema data model, DAX measures, Phase 3 dashboard pages
- 1-page insight summary (PDF/Word) for the VP: key findings + 3 concrete recommendations